

TUTORIAL 9: 16.1 AND 16.2

16.1

Find the gradient vector field of f .

(21) $f(x, y) = \tan(3x - 4y)$.

(22) $f(s, t) = \sqrt{2s + 3t}$.

(24) $f(x, y, z) = x^2 y e^{y/z}$.

(25) Find the gradient vector field ∇f of f and sketch it. $f(x, y) = \frac{1}{2}(x - y)^2$.

16.2

Evaluate the line integral, where C is the given curve.

(5) $\int_C (x^2 y + \sin x) dy$, where C is the arc of the parabola $y = x^2$ from $(0, 0)$ to (π, π^2) .

(13) $\int_C x y e^{yz} dy$, $C : x = t, y = t^2, z = t^3, 0 \leq t \leq 1$.

(14) $\int_C z dx + x dy + y dz$, $C : x = t^2, y = t^3, z = t^2, 0 \leq t \leq 1$.

(15) $\int_C z^2 dx + x^2 dy + y^2 dz$, C is the line segment from $(1, 0, 0)$ to $(4, 1, 2)$.

(20) Evaluate the line integral $\int_C F \cdot dr$, where $F(x, y, z) = (x + y^2)i + xzj + (y + z)k$.
 $r(t) = t^2 i + t^3 j - 2tk, 0 \leq t \leq 2$.